Biomechanical Analyses of Bodyweight Unilateral Lower Limb Exercise Tasks – Comparison of Common Squatting and Lunging Movements

PREVIEW

by

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A thesis proposal submitted in conformity with the requirements
for the degree of Master of Science

Department of Exercise Sciences
University of Toronto

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Abstract

Bodyweight unilateral lower limb exercise tasks (BULLETs) are used in movement assessment and (re)training contexts. Since kinematic characteristics of individual BULLETs are different, it was hypothesized that lower extremity biomechanical loading patterns would differ between BULLETs. Thirty-two participants performed bodyweight forward lunges, backward lunges, split squats, single-leg squats, and double-leg squats. Body segment kinematics and ground reaction forces were used to quantify body-size normalized, lower limb net joint moments. Peak and average hip, knee, ankle, and total support moments differed across BULLETs ($p < 0.05$). Sex-based differences were observed for peak and average support moment, and peak knee moment ($p < 0.05$). Characterizing BULLETs based on joint moment magnitudes can be used to improve the sensitivity and specificity of movement assessments and to design movement (re)training programs. Further work is needed to address cause(s) of between-sex differences in joint moment magnitudes given that body size differences were not explanatory.

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Acknowledgments

This manuscript is dedicated to my grandparents:

John (Jack) and Shirley MacFarlane

I would not be the person I am today without their love, support, influence and inspiration.

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1 Introduction

Resistance training is a fundamental intervention used in the health care and fitness industries, and is recommended to the general population as part of a healthy, active lifestyle (CSEP, 2014; Garber et al., 2011). Bodyweight unilateral lower limb exercise tasks, referred to herein as BULLETs (e.g., lunge, split squat, single-leg squat), are commonly used in exercise prescription to elicit specific adaptations such as increasing lower limb strength and muscle mass (Baechle & Earle, 2008; Bizzini, Junge, & Dvorak, 2007; Tsatsouline, 2003). Furthermore, these tasks are included in clinical assessments in an attempt to determine an individual's risk of sustaining an injury based on the patterns of movement coordination and control exhibited during task execution (M. A. Clark & Lucett, 2010; Comerford & Mottram, 2012; Cook, Burton, Kiesel, Rose, & Bryant, 2010; Kritz, 2012; Mischiati et al., 2015).

If the human body is modeled as a complex mechanical linkage, the kinematics and kinetics of its motion are intrinsically linked (Zajac, 1993). Since the kinematic characteristics of individual BULLETs are inherently different, it is therefore likely that the biomechanical loading patterns amongst lower limb joints differ between tasks. For example, between-task differences in the configuration of body segments suggests that whole-body center of mass (COM) trajectories would differ across BULLETs and result in different ground reaction force profiles to maintain postural stability; this would affect the external joint moment of force and associated (i.e., equilibrating) muscle force patterns. Moreover, muscle lengths and orientations would vary between tasks due to different body segment and joint angles, causing differences in the muscle moment arms, contraction velocities, and thus their force- and moment-producing capabilities.

Although acknowledged that the addition of external resistance or variations in movement speeds could affect the abovementioned relationships in complex ways (Frost, Beach, Callaghan, & McGill, 2013), it is nevertheless beneficial to investigate biomechanical differences between BULLETs because performing exercises without added resistance and at low speeds often constitutes a baseline condition for novice or uninjured trainees. Further, bodyweight tasks are commonly incorporated as part of a battery of tasks in qualitative movement assessments (M. A. Clark & Lucett, 2010; Comerford & Mottram, 2012; Cook et al., 2010; Kritz, 2012). In this specific case, understanding how similar or different these tasks are

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in lower limb loading may enhance the rationale for their inclusion or exclusion, which in turn could enhance the validity, specificity or sensitivity of assessments that incorporate one or more of these movement tasks.

Lower limb internal net joint moments have been used to great extent in biomechanical research as a simple but crude means to quantify the total outcome effect of all internal force-producing and -bearing structures surrounding a joint (Andriacchi, Andersson, Fermier, Stern, & Galante, 1980; S. P. Flanagan & Salem, 2008; Gullett, Tillman, Gutierrez, & Chow, 2009; Mathiyakom, McNitt-Gray, Requejo, & Costa, 2005; Riemann, Congleton, Ward, & Davies, 2013; Salem, Salinas, & Harding, 2003; Singh, Yack, Francis, & Janz, 2015; Winter, 1980). Knowledge of the lower limb net joint moments associated with performing BULLETs, in particular, could aid practitioners in the design and administration of these exercises (Herbert, Moore, Moseley, Schurr, & Wales, 1993). As opposed to primarily manipulating external resistance and/or movement speed of a given exercise, different BULLETs could also be selected to progress/regress the crude biomechanical demands imposed on the basis of the associated net joint moment magnitudes. This would be especially useful for individuals with limited experience (e.g., novice trainees) and/or those who are not physically capable to safely execute exercises at high movement speeds and/or against additional external resistance (e.g., sedentary or previously injured trainees). Additionally, using different BULLETs as a means to alter biomechanical demands could provide a method of progressive overload when equipment and coaching are unavailable. Having greater control over progressive overload and a further understanding of the imposed lower limb demands during such exercise tasks could also prove useful in the prevention of injuries, as the application and rate of application of excessive load without adequate adaptation or recovery is suggested to be a primary cause of training related injuries (Blanch, 2014; Kumar, 2001; McGill, 1997).

Quantifying and comparing the lower limb net joint moments across BULLETs can bring practitioners a step closer to integrating more biomechanical logic into planning exercise programs (i.e., periodization). Current practice focuses more on manipulating external loading parameters (e.g., added mass, system mass, etc.), with less regard for the associated joint loading demands (e.g., net joint moments). In cases where specific musculoskeletal exercise adaptations are desired, using net joint moment magnitudes to inform exercise prescription and progression is arguably more ecologically valid. Given that the majority of joint motion during

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BULLETs is confined to the sagittal plane, net joint extension (hip and knee) and plantarflexion (ankle) moment magnitudes could be considered for each of the primary lower limb joints individually as a means to crudely quantify the task demand. Knowledge of these specific joint moment magnitudes would support a practitioner/clinician, for example, during a rehabilitation or task-transfer scenario where the attentional focus and desired outcomes are targeted locally at a single joint (e.g., improved hip extensor strength). In cases when the objective is to elicit a more global adaptation (e.g., lower limb concentric/eccentric strength improvement), the practitioner could use knowledge of a combination of moments in the exercise planning process. For example, the "support moment" can be calculated by adding the net joint extension (hip and knee) and plantarflexion (ankle) moments (Winter, 1980), and used a valid measure of total lower limb mechanical demand (S. Flanagan & Salem, 2005).

One important consideration when using bodyweight exercises for the abovementioned purposes is that segment masses and lengths differ between individuals – especially between men and women (Sizer & James, 2014) – which would affect net joint moment magnitudes. It is common practice to normalize net joint moment magnitudes by dividing them by body mass and/or size; this is done to account for differences in anthropometry so that inter-individual comparisons can be made (Moisio, Sumner, Shott, & Hurwitz, 2003). However, likely related to sexual dimorphism in the human musculoskeletal system (Sizer & James, 2014), even when biomechanical data are normalized with respect to body size and shape characteristics, between-sex differences still exist in single-leg squat and lunge mechanics (Bolga, Cook, Hogarth, Scott, & West, 2014; Dwyer, Boudreau, Mattacola, & Uhl, 2010; Graci, Van Dillen, & Salsich, 2012; Nakagawa, Moriya, Maciel, & Serrao, 2012; Zeller, McCrory, Kibler, & Uhl, 2003). Thus, it is important to consider if between-sex differences exist in the lower limb net joint moments of various BULLETs, as interpretation and application of results may vary between men and women.

To date, no research has directly compared the lower limb net joint moments and support moment between the various BULLETs performed for use in the aforementioned scenarios. Thus, the primary objective of this research was to quantify lower limb net joint moments during commonly prescribed BULLETs, and to compare their magnitudes, and a summation of these magnitudes between BULLETs. A secondary objective was to make between-sex

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comparisons in these quantities to determine if net joint moment magnitudes differ between men and women when normalized for body size.

1.1 Questions

1.1.1 Question 1 – Joint Kinetics

Do mean and peak lower limb net joint extension (hip and knee) and plantarflexion (ankle) moments, and their algebraic summation (i.e., total support moment), differ between four common BULLETs (i.e. single-leg squat, split squat, forward lunge, backward lunge) and a baseline non-BULLET task (i.e., double-leg squat)?

1.1.2 Question 2 – Sex Differences

Are any between-BULLET differences in the abovementioned kinetic quantities dependent on the sex of the performer when body-size and mass are accounted for by dividing net joint moment magnitudes by body mass × leg length?

1.2 Hypotheses

1.2.1 Null Hypothesis 1

There will be no difference in the mean or peak net joint extension (hip and knee), plantarflexion (ankle), or total support moment between the BULLETs.

1.2.2 Null Hypothesis 2

Any between-BULLET differences in the abovementioned kinetic quantities will be consistent between men and women.

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2 Review of Literature

The objective of this section is three-fold. First, a summary of the current state of knowledge and gaps that exist with respect to selected BULLETs is provided. Second, an overview of how practitioners currently use BULLETs in exercise paradigms is offered as way to demonstrate the relevance of the work undertaken. Third, justification of the methodology employed is provided.

BULLET is an acronym used in this thesis for a collection of "bodyweight unilateral lower limb exercise tasks", which include kinematically asymmetrical, ground-based, free-standing movements whose motion primarily involves flexion/extension at the hip and knee and dorsiflexion/plantarflexion at the ankle. 'Bodyweight' implies that no additional external resistance is added to the body, only an individual's weight (i.e., body mass × acceleration due to gravity) offers the resistance sustained. The exercise tasks investigated herein are the forward lunge (FL), backward lunge (BL), split squat (SS), and single-leg squat (SLS; Appendix C). The double-leg squat [DLS] is used as a baseline task for comparison. Rationale for the selection of these BULLETs will be provided in justification of methodology.

All BULLETs consist of the following positions and motions in sequential order: an initial starting position; an eccentric (downward) phase of movement; a transition phase from eccentric to concentric also known as the bottom position; a concentric (upward) phase of movement; and ending with the final position which is identical to the initial position. For the purposes of this study, hands were placed on the pelvis (i.e., hands on "hips") in order to minimize variations in upper limb positioning and its influence on kinematics and COM motion. The dominant leg was considered the "stance-leg" moving forward. The specific kinematic variation and a summary of the available literature to date for each task will be explained subsequently in its respective section.

2.1 Single-Leg Squat (SLS)

The SLS begins with an initial position of upright standing with feet flat on the ground, hip- to shoulder-width apart. The hip and knee joints are initially extended while the ankle joint remains in a neutral position. The non-stance leg is then elevated slightly so that it does not come in contact with the ground at any point during the movement task. Movement is initiated

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by lowering the COM towards the ground via (dorsi)flexion about the hip, knee and ankle (i.e., triple flexion) while attempting to maintain a neutrally-oriented thoracolumbar spine and permitting minimal mediolateral motion of the knee joint. The transition point or bottom of movement occurs once a maximum controllable squat depth is reached. All three lower limb joints then subsequently extend to produce upward COM motion. The movement ends when the final position is reached, which is identical to the initial position. For the purposes of this study, the participant was instructed to maintain postural stability without allowing the elevated foot to touch the ground for the duration of the movement task.

Variations in the position of the elevated foot in the SLS do exist (i.e., foot in front of stance leg, foot behind the stance leg). One unpublished pilot study by the author has shown that while stance leg kinematics and inter-joint load sharing remain largely unaffected by elevated foot position, body mass-normalized sagittal net joint moments differ at the hip and knee (Chapman et al., 2014). Although this pilot study was statistically underpowered, the results were confirmed in an appropriately powered study (A. Khuu, Foch, & Lewis, 2016), and thus provides rationale to control position of the elevated foot during SLS performance.

During quantitative kinetic and kinematic analysis, all joint angles, moments and ground reaction forces can be reliably measured between sessions within a given day and within a given week for the SLS (Alenezi, Herrington, Jones, & Jones, 2014). These measurement methods have been widely employed to justify its use as a qualitative movement screen, more so than the other BULLETs studied herein. Visual observation of the SLS has shown to be reliable between raters for both inferring joint angle/segment positions (Ageberg et al., 2010; Edmondston et al., 2013; Poulsen & James, 2011) and qualitative scoring scales (Almangoush, Herrington, & Jones, 2014; Crossley, Zhang, Schache, Bryant, & Cowan, 2011; Junge, Balsnes, Runge, Juul-Kristensen, & Wedderkopp, 2012). This reliability increases with education and experience (Weeks, Carty, & Horan, 2012). Using visual observation of the SLS in movement screens is supported as it sensitive to assessing knee valgus or medial knee deviation (Ugalde, Brockman, Bailowitz, & Pollard, 2014), which has been shown to be a risk factor for lower limb non-contact knee injuries (K.R. Ford, Myer, & Hewett, 2003; Hewett et al., 2005).

SLS performance is also related to hip muscle function (Crossley et al., 2011; Hollman, Galardi, Lin, Voth, & Whitmarsh, 2014; Stickler, Finley, & Gulgin, 2015). People who exhibit

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less medial knee displacement during the SLS have faster onset of hip abductor activation and stronger hip muscles. Hip muscle function is greatly compromised in people with anterior knee pain (Nakagawa et al., 2012; Piva, Goodnite, & Childs, 2005; Prins & van der Wurff, 2009). Individuals who have patellofemoral pain syndrome (PFPS) exhibit approximately 20% less strength in hip abduction and external rotation compared to healthy controls, and show greater knee valgus (Levinger, Gilleard, & Coleman, 2007; Nakagawa et al., 2012), in addition to displaying limited ankle dorsiflexion (Mauntel et al., 2013). This would suggest that all primary joints of the lower limb need to be considered when practitioners interpret SLS performance. In addition to exhibiting greater knee valgus and less ankle dorsiflexion than their unaffected counterparts, individuals with PFPS also show greater ipsilateral trunk lean, contralateral pelvic drop, and hip adduction. A more upright trunk position during SLS has also been shown to increase the force and strain of the anterior cruciate ligament (Kulas, Hortobagyi, & DeVita, 2012),

Further evidence from an intervention study supports the link between hip musculature and performance during the SLS (Willy & Davis, 2011). Specifically, healthy women were assessed with excessive hip adduction using instrumented motion capture during running and placed into an intervention group. This group completed a hip movement and strengthening program focused on the abductors and external rotators that consisted of progressions from side lying to standing, from supported to non-supported, and from unloaded to added resistance. This program was performed 3 times per week for 6 weeks and a control group comprised of participants who did not display excessive hip adduction during the SLS were included. While the intervention group showed no changes in hip adduction during running, they did significantly decrease hip adduction, hip internal rotation and contralateral pelvic drop during SLS task. The training intervention was designed to improve hip muscle capacity, and resulted in improvements in kinematics of SLS performance, but did not influence the kinematics of running performance.

There is more SLS literature that is predominantly frontal plane focused, but is not relevant to the study at hand and therefore not reported. Surprisingly, with the vast amount of research on the kinematics and muscle activation of SLS performance, only one study has quantified the net joint moments of the hip, knee and ankle during this exercise task (A. Khuu et al., 2016). The results of this study showed that changing the elevated foot/leg position during a SLS changed

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the net joint moments observed, in addition to other kinetic and kinematic variables. While this provides published values to compare the results herein against, the authors only extracted the moment values at two discrete kinematic time points, 60 degrees knee flexion and peak knee flexion. For the purpose herein, the peak and average net joint moments were examined to understanding the magnitude of maximum loading during the SLS and the average total loading for the duration of the task. While there is some evidence to suggest the former may occur at peak flexion angles during squatting tasks (Cotter, Chaudhari, Jamison, & Devor, 2013; Scaglioni-Solano, Ferris, Dunn, & Salem, 2005), the latter does not relate to these discrete time points. Therefore, the current study will add to the body of literature of net joint moment responses observed during SLS exercise tasks. Furthermore, to date no studies have compared the net joint moments of SLS to other BULLETs nor DLS in order to determine and differences and potentially categorize the observed magnitudes relative to one another.

2.2 Forward Lunge / Backward Lunge / Split Squat (FL / BL / SS)

The FL and BL tasks involve the same initial position of upright standing with feet hip- to shoulder-width apart. Movement is initiated with the stance-leg stepping and COM moving down and forward (FL), or the non-stance leg stepping and COM moving down and backward (BL). Similar to SLS, an attempt is made to control frontal plane knee motion and maintain a neutral spine throughout the duration of the exercise task. The transition phase, or bottom of the movement, occurs either when the back knee (nearly) contacts ground, and/or when the front knee reaches a specific joint angle, usually at or around 90 degrees (Graham, 2002; Keogh, 1999).

The SS slightly differs in initial position as it begins with feet displaced in the anterior-posterior direction at lunge step-length, with the stance leg being the anterior leg. This results in the front-leg hip being flexed and the rear-leg hip is extended, while an attempt is made to maintain the trunk in an upright position. However, both knees are fully extended and this is used as the cue for initial and final position being achieved. Feet remain in place while COM motion is initiated downward to the same transition phase as the lunge, followed by upward COM motion returning to the starting position (Schütz et al., 2014).

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